

Triasha Sarkar

Machine Learning Engineer | Ex-Rolls-Royce · United States · tsarkar34@gatech.edu · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Machine learning engineer with aerospace systems experience building evaluation pipelines, retrieval systems, surrogate models, and deployment evidence. I work across data contracts, model evaluation, reproducible experiments, and production boundaries. My focus is making ML results measurable, reviewable, and useful to engineering teams.

EXPERIENCE

Machine Learning Engineer — Rolls-Royce Jul 2023 – Apr 2025

- Built machine-learning and predictive-maintenance workflows for aircraft-engine sensor data.
- Developed Python pipelines for 1 Hz and 10 Hz multivariate time-series data, including cleaning, feature engineering, anomaly detection, and model evaluation.
- Worked with Lifecycle Engineers and domain experts to understand failure modes and turn model outputs into maintenance insights.
- Used Azure Databricks for repeatable analysis of large engineering datasets and contributed to solutions supporting earlier issue detection.

Graduate Research Assistant — Aerospace Systems Design Laboratory, Georgia Tech May 2025 – Aug 2026

- Built a failure-detection workflow using 15,120 simulated rocket-motor time-series cases, reaching 96.8% accuracy and 0.986 ROC AUC on unseen geometries.
- Worked on the Delta Air Lines-sponsored HERO project, exploring retrieval and RAG methods for safety-related information.
- Developed surrogate models for the GREEN TEA project, connecting lifecycle, economic, and transportation data for faster engineering analysis.
- Applied Python, scikit-learn, statistical modeling, feature engineering, clustering, and model evaluation with a focus on reproducibility and failure analysis.

Project EAGLE — Vehicle Grand Challenge with RIC Enterprise, Georgia Tech Sep 2025 – Present

Worked on Project EAGLE (Eco-friendly Airborne Global Luxury Ekranoplan), a two-semester industry-sponsored research project. Developed a sizing and synthesis tool for a one-of-a-kind vehicle, and developed the passenger-accommodation layout and cabin geometry in AutoCAD while modifying and developing vehicle requirements.

Project GREEN TEA — System of Systems Grand Challenge, Georgia Tech Sep 2025 – Present

Worked on Project GREEN TEA (Generalized Residue to Energy Evaluation Network and Techno-Economic Analysis), an industry-sponsored project with the U.S. Endowment for Forestry and Communities. Developed the architecture and framework for a techno-economic tool analyzing sustainable aviation fuel and bioenergy from forest residues, including jobs, CO₂ impacts, a GREET life-cycle-analysis surrogate model, and a forest-health impact model.

Intern — STAR Orbitals Aug 2021 – Oct 2021

Interned at STAR Orbitals from Aug 2021 to Oct 2021, contributing to static-test-pad development for solid rocket motors, client timelines, project planning, Arduino avionics, SolidWorks, and Autodesk Fusion 360 mechanical design.

Intern — Rolls-Royce DataLabs May 2021 – Jul 2021

Worked on aerospace data projects involving predictive analytics, anomaly detection, model-based engineering, exploratory analysis, feature engineering, and time-series sensor data in a safety-critical environment.

Summer Intern — Bangalore Aircraft Industries (BAIL) Jun 2019 – Jul 2019

Worked on material testing, stress analysis, and structural testing using NASTRAN-PATRAN.

SELECTED ENGINEERING PROJECTS

AeroRAG-X — Retrieval and agent evaluation [GitHub](#)

Built a provenance-aware RAG evaluation system with BM25, dense retrieval, hybrid reranking, citation checks, safe agent degradation, and held-out QASPER/SciFact evaluation. Measured retrieval behavior from 10K to 1M NASA-text snapshots rather than assuming scale preserved ranking quality.

NewsLens — Stateful news retrieval service [GitHub](#)

Built a FastAPI news-retrieval service with Go ingestion, Kafka, PostgreSQL, Docker/Kubernetes deployment paths, and

reproducible MIND evaluation. The selected content-plus-fallback model reached NDCG@10 0.3664; a weaker learned ranker was rejected under the frozen comparison.

IntegrityBench — Integrity and safety evaluation[GitHub](#)

Built an integrity-evaluation and deployment workflow measuring macro F1, false acceptance, false rejection, calibration, policy citation accuracy, and latency. Added public evidence receipts, CUDA evaluation paths, release gates, monitoring hooks, and an AWS reference deployment plan.

EdgeGenBench — Edge inference and flight-anomaly evaluation[GitHub](#)

Built an ONNX/QNN deployment-evidence pipeline and evaluated flight anomalies on 17,780 aircraft-disjoint approaches. The model reached 94.96% accuracy but 0.7380 macro F1, so the release remains blocked instead of relying on accuracy alone.

SELECTED ENGINEERING PROJECTS (CONTINUED)

AeroSynth-Eval — Generative-model evaluation[GitHub](#)

Built preference, calibration, transfer, downstream-utility, grouped-uncertainty, and human-alignment evaluation tooling. In a ten-seed AGDD study, mixed training improved macro F1 from 0.3881 to 0.4419 but reduced crack recall, so the release gate remains blocked.

Equity-Backtest — Leakage-aware research pipeline[GitHub](#)

Built a monthly cross-sectional backtest with point-in-time membership validation, no-look-ahead controls, turnover, capacity constraints, and multiple-testing records. The pipeline explicitly withholds return claims pending licensed CRSP constituents and delisting returns.

Surrogate Model Learning — Robust regression[GitHub](#)

Built Gaussian-process and regression workflows for public engineering datasets with grouped holdouts, operating-domain warnings, uncertainty checks, and data-hashed repeated robustness reports. Airfoil robustness across ten grouped splits reached mean R^2 0.8662.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, MATLAB, Bash

ML: PyTorch, scikit-learn, Gaussian processes, regression, retrieval, ranking, evaluation, uncertainty, time series

Systems: Docker, Kafka, Postgres, AWS, CI, ONNX, QNN, REST APIs, observability, reproducibility contracts

Engineering: requirements management, systems architecture, verification and validation, risk analysis, model cards, technical documentation

RESEARCH AND ENGINEERING FOCUS

- Scientific machine learning and surrogate modeling for aerospace and energy systems.
- Evaluation of retrieval, generation, safety, citation support, and model integrity.
- Reliable ML systems with explicit data, deployment, and evidence boundaries.

EARLIER AEROSPACE PROJECTS

- **GKN Aerospace hybrid aircraft design:** thermal-management integration for a zero-emission demonstrator using hybrid, electric, and hydrogen system concepts.
- **Supersonic business-jet thesis:** studied primary-nozzle geometry and noise-mitigation constraints for take-off and landing.
- **Bird-strike structural analysis:** used ANSYS to compare nose-cone material response under impact loading.

EDUCATION

M.S., Aerospace Engineering — Georgia Institute of Technology Aug 2026

M.Sc., Aerospace Vehicle Design — Cranfield University Mar 2022 – Feb 2023

B.Tech., Aerospace Engineering — SRM Institute of Science and Technology Jul 2017 – Jun 2021